Network and Connectivity Troubleshooting Guide

This guide covers all network and connectivity issues: Wi-Fi, VPN, Ethernet, DNS, and proxy settings.

1. Wi-Fi Connection Issues

1.1 Cannot Connect to Corporate Wi-Fi (CorpNet)

The corporate Wi-Fi uses WPA2-Enterprise authentication with Active Directory credentials.

- Step 1: Forget the network. Go to Wi-Fi Settings, select CorpNet, and choose Forget Network.
- Step 2: Re-scan and select CorpNet again.
- Step 3: Enter your AD credentials: username@company.com and your AD password.
- Step 4: Accept the security certificate if prompted.
- Step 5: Flush DNS if still failing. Open Command Prompt as Admin and run: ipconfig /flushdns

INFO: Corporate Wi-Fi uses WPA2-Enterprise. Personal Wi-Fi passwords will NOT work on CorpNet.

1.2 Wi-Fi Connected but No Internet

- Run ipconfig (Windows) or ifconfig (Mac/Linux). Valid IP is 10.x.x.x or 192.168.x.x.
- If you see 169.254.x.x, your DHCP lease failed. Disconnect and reconnect.
- Test DNS: ping 8.8.8.8. If ping succeeds but websites fail, DNS is broken.
- Set DNS to 10.0.0.1 (Primary) and 10.0.0.2 (Secondary) in network adapter settings.

1.3 Slow Wi-Fi

- Switch to CorpNet-5G (5GHz band) for faster speeds if your device supports it.
- Move closer to the nearest access point. Floor maps on the intranet show AP locations.
- Restart your network adapter via Device Manager: right-click adapter, Disable then Enable.

2. VPN - Remote Access

2.1 Installing Cisco AnyConnect VPN

- Windows: Download from https://vpn.company.com/anyconnect-win.exe and run as Administrator.
- macOS: Download from https://vpn.company.com/anyconnect-mac.dmg and run the installer.
- Linux: Download from https://vpn.company.com/anyconnect-linux.tar.gz, extract, and run install.sh as root.

```
VPN Server: vpn.company.com Group: Corporate-Employees Username: your AD
username (without @company.com) Password: your AD password
```

2.2 VPN Connection Errors

- Certificate Invalid: Sync your system clock. VPN certificates are time-sensitive.
- Authentication Failed: Reset your AD password at https://password.company.com.
- Unable to establish VPN: Check if local firewall blocks port 443 or 1194.
- No valid license: Contact IT - your account may need VPN group membership added.

INFO: VPN sessions auto-disconnect after 8 hours of inactivity. Re-authenticate to reconnect.

2.3 VPN Gateways by Region

- US East: vpn-east.company.com
- US West: vpn-west.company.com
- EU London: vpn-eu.company.com
- APAC Singapore: vpn-apac.company.com
- Always connect to the nearest gateway for best performance.

3. Ethernet - Wired Connection

3.1 No Network on Ethernet

- Try a different Ethernet cable and a different wall jack port.
- Check the port LED: solid green means link established; blinking means traffic.
- Run ipconfig /all and verify you have a valid IP (not 169.254.x.x).
- Disable and re-enable the network adapter in Device Manager.
- If using a docking station: connect directly to the laptop HDMI port to isolate.

INFO: All Ethernet ports require 802.1X authentication. Ensure your machine is domain-joined.

4. DNS and Proxy Settings

4.1 Corporate DNS Configuration

```
Primary DNS: 10.0.0.1 Secondary DNS: 10.0.0.2 DNS Suffix: company.com Proxy
Server: proxy.company.com:8080
```

4.2 Browser Proxy Setup

- Chrome/Edge: Settings, System, Open proxy settings, Use automatic configuration script.
- Proxy PAC URL: http://proxy.company.com/proxy.pac
- Firefox: Settings, Network, Manual proxy, HTTP: proxy.company.com, Port: 8080.
- For CLI tools: export https_proxy=http://proxy.company.com:8080

5. Network Access Requests

5.1 Requesting a Blocked Website Unblock

- Go to helpdesk.company.com, New Ticket, Category: Network, Subcategory: Firewall Exception.
- Provide: URL to unblock, business justification, and manager approval email.
- SLA: Reviewed within 2 business days.

5.2 Requesting a Static IP Address

- Submit via IT helpdesk portal. Provide device MAC address, business reason, and location.
- Static IPs are assigned from the range 10.50.0.1 to 10.50.0.254.
- SLA: 3 business days for provisioning.